Generated by Scopus AI, Sat Oct 10 2026

Neural networks for DNA sequence analysis

Neural Network Models for DNA Sequence-Based Disease Prediction: Methods, Challenges, and Clinical Translation

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Neural Network Architectures	Hybrid CNN-RNN, transformer-based genome language models, and NAS-optimized models are state-of-the-art for DNA sequence analysis.	1 2 3 4
Hybrid & Multimodal Integration	Combining CNNs, RNNs, GNNs, GANs, and multi-modal data improves predictive accuracy and robustness.	5 6 7 8
Interpretability & Clinical Applicability	SHAP, LIME, Grad-CAM, and integrated gradients are critical for clinical trust and adoption.	9 10 11 12
Uncertainty Quantification	Bayesian neural networks and ensemble methods enhance decision-making and generalizability.	13 14 15 16
Data Challenges & Integration	Data imbalance, noise, limited labels, and multi-omics integration addressed by self-supervised learning and federated learning.	17 18 19 20

Direct Answer

Your ideathon project, which aims to predict disease risk from DNA sequences using neural networks, is at the intersection of advanced genomics and AI. To move forward, investigate state-of-the-art neural models such as hybrid CNN-RNN architectures, transformer-based genome language models (e.g., DNABERT), and techniques in neural architecture search (NAS) tailored to genomic data. Key research papers include reviews and studies on k-mer versus transformer-based embedding techniques [1], hybrid models combining GNNs and GANs [5], and the latest in self-supervised learning approaches like Self-GenomeNet [18]. Additionally, focus on integrating robust uncertainty quantification (via Bayesian neural networks) and interpretability (using SHAP, LIME, or Grad-CAM) to ensure clinical relevance. Consider exploring standard genomic datasets such as those available on NCBI, ENCODE, and the HMDD databases for training and benchmarking, and review papers on multi-omics integration, as described in recent surveys [6] [21].

Study Scope

- **Time Period:** Recent advances (last 5 years) with historical context for foundational models.
- **Disciplines:** Bioinformatics, genomics, machine learning, clinical medicine.
- **Methods:** Meta-analysis of neural network architectures, benchmarking studies, interpretability frameworks, and clinical translation research.

Assumptions & Limitations

- **Assumptions:** Neural networks can capture complex sequence patterns and gene-gene interactions; interpretability and uncertainty quantification are essential for clinical adoption.
- **Limitations:** Lack of universally standardized guidelines, limited external validation, and challenges in integrating multi-omics and clinical data.

Suggested Further Research

- Develop modular pipelines integrating interpretability and uncertainty quantification for actionable clinical reports.
- Standardize external validation protocols and guidelines for DNA-based disease prediction models.
- Advance multimodal integration and federated learning for robust, privacy-preserving genomic analysis.

1. Introduction

Background and Motivation

The application of neural networks to DNA sequence analysis for disease prediction is revolutionizing personalized medicine. By leveraging deep learning, researchers aim to decode the complex relationships between genetic variants and disease risk, enabling early diagnosis and tailored interventions. The rapid growth in genomic data, coupled with advances in AI, has made it feasible to predict future diseases from DNA sequences, identify pathogenic variants, and generate actionable clinical reports [17] [22] [23]. This approach promises to bridge the gap between genotype and phenotype, offering unprecedented opportunities for healthcare.

Synthesis: The convergence of genomics and AI is driving innovation in disease prediction, with neural networks offering the ability to model intricate sequence patterns and gene-gene interactions. However, challenges in interpretability, data quality, and clinical translation remain significant [17] [22] [23].

2. Neural Network Architectures for DNA Sequence Analysis

Convolutional and Recurrent Neural Networks

- **CNNs:** Capture spatial motifs in DNA sequences, effective for local feature extraction and classification tasks [4] [24].

- **RNNs (LSTM, GRU):** Model sequential dependencies, suitable for temporal and long-range interactions in genomic data [25] [26] [27].
- **Hybrid CNN-RNN:** Combine spatial and sequential modeling, outperforming standalone models in binding specificity and disease prediction [26] [27] [28].

Comparative Table: CNN vs. RNN vs. Hybrid Models

Model Type	Strengths	Limitations	Use Cases
CNN	Local motif detection	Limited long-range modeling	DNA region classification
RNN (LSTM/GRU)	Sequential dependencies	Slow training, vanishing gradients	Temporal genomics, time-series
Hybrid CNN-RNN	Both spatial & sequential	Reduced interpretability	Disease risk prediction

Synthesis: Hybrid architectures are increasingly favored for their ability to capture both local and global sequence features, though interpretability remains a challenge [26] [27].

Transformer-Based and Genome Language Models

- **Transformers (e.g., DNABERT):** Model long-range dependencies and global context, outperforming k-mer embeddings in genomic signal prediction [1] [2] [29].
- **Genome Language Models (gLMs):** Self-supervised pretraining on raw nucleotide sequences enables functional genomics and variant effect prediction [2] [30].
- **Multimodal Integration:** Foundation models integrate DNA, transcriptomics, and spatial data for comprehensive disease prediction [29].

Synthesis: Transformer-based models are setting new benchmarks in genomic feature extraction and disease prediction, with multimodal integration enhancing accuracy and interpretability [1] [2] [29].

Graph Neural Networks and Hybrid Models

- **GNNs:** Capture gene-gene interactions and network structure, effective for modeling epistasis and disease associations [5] [31].
- **GANs:** Augment training data and improve robustness, especially when combined with GNNs [5] [31].

- **Hybrid Fusion:** Combining GNNs, transformers, and generative AI enables multimodal data integration and improved disease classification [32].

Synthesis: Hybrid GNN-GAN models are promising for capturing complex biological interactions, though challenges in interpretability and clinical translation persist [5] [31] [32].

Neural Architecture Search (NAS) and Model Optimization

- **NAS:** Automates discovery of optimal architectures, balancing performance and computational efficiency [3] [33].
- **Training-Free NAS:** Reduces computational cost, making NAS accessible for genomic sequence analysis [34].
- **Multi-Objective Optimization:** Balances accuracy, interpretability, and resource constraints [35] [36].

Synthesis: NAS is increasingly used to tailor neural network architectures for genomic data, enabling efficient and scalable model development [3] [33] [34].

3. Biological and Computational Challenges

Modeling Gene-Gene Interactions (Epistasis)

- **Epistasis Detection:** Neural frameworks (e.g., EpiDetect, EpiCID) and explainability algorithms identify SNP-SNP interactions and biologically relevant pathways [37] [38].
- **Optimization Algorithms:** Brain Storm, Sparrow Search, and Ant Colony Optimization improve detection power and scalability [39] [40] [41].
- **Reinforcement Learning:** Models epistasis as a Markov Decision Process, refining gene selection and interaction discovery [42].

Synthesis: Advanced neural and optimization frameworks are enhancing the detection of complex gene-gene interactions, critical for accurate disease prediction [37] [38].

Data Imbalance, Noise, and Privacy

- **Data Imbalance:** Cost-sensitive learning, attention mechanisms, and data augmentation address imbalance in genomic datasets [43] [44].
- **Noise Reduction:** Multi-view models and confident learning mitigate label noise and improve robustness [45] [46].
- **Privacy-Preserving AI:** Federated learning and homomorphic encryption enable secure collaborative training on sensitive genomic data [19] [47].

Synthesis: Addressing data quality and privacy is essential for reliable disease prediction, with advanced AI techniques providing scalable solutions [19] [43] [45].

Interpretability and Explainability

- **Interpretability Methods:** SHAP, LIME, attention mechanisms, and saliency maps improve transparency and clinical trust [48] [49] [50].
- **Bio-Centric Interpretability:** Integrating biological domain knowledge enhances model plausibility and generalization [51].
- **Explainable Multi-Task Models:** EMDIM and ExplaiNN provide transparent predictions for multi-task DNA sequence analysis [49] [52].

Synthesis: Interpretability is a cornerstone for clinical adoption, with explainable AI methods bridging the gap between predictive accuracy and actionable insights [48] [49] [50].

Ethical, Regulatory, and Usability Considerations

- **Ethical Challenges:** Data privacy, reproducibility, and responsible AI deployment are critical for clinical translation [17] [53].
- **Regulatory Barriers:** Lack of standardized guidelines limits model validation and adoption [47] [53].
- **Usability:** User-friendly interfaces and clinician engagement are necessary for widespread adoption [17].

Synthesis: Ethical and regulatory frameworks must evolve alongside technical advances to ensure safe and effective clinical deployment [17] [47] [53].

4. Datasets and Benchmarks

Public Genomic and Multi-Omics Datasets

- **NCBI, HMDD, panX:** Widely used for training and benchmarking disease prediction models, with labeled DNA sequences and disease phenotypes [21] [54] [55].
- **Multi-Omics Resources:** Integrate transcriptomics, proteomics, and clinical data for comprehensive modeling [56] [57].

Synthesis: Large-scale, annotated datasets are foundational for model training and evaluation, with multi-omics integration enhancing predictive power [21] [54] [55].

Benchmarking Studies and Evaluation Metrics

- **Comparative Studies:** CNN, RNN, Bayesian, and hybrid models benchmarked across diseases (HIV, cancer, cardiovascular) [58] [59] [60].
- **Evaluation Metrics:** Accuracy, AUC, F1, interpretability scores, and external validation are standard [59] [60] [61].

Synthesis: Rigorous benchmarking and diverse evaluation metrics are essential for assessing model performance and clinical readiness [58] [59] [60].

Multi-Modal Data Integration

- **Integration Strategies:** Combine DNA, clinical, imaging, and multi-omics data for improved disease prediction [6] [62] [63].
- **Graph Neural Networks:** Enable heterogeneous data fusion and modality importance analysis [62].

Synthesis: Multi-modal integration is driving advances in disease prediction, capturing complex biological and clinical interactions [6] [62] [63].

Self-Supervised and Semi-Supervised Learning

- **Self-GenomeNet:** Leverages unlabeled genomic data, improving performance with fewer labeled samples [18].
- **Semi-Supervised Methods:** Enhance classification accuracy and generalization in data-scarce settings [64] [65].

Synthesis: Self-supervised learning is a powerful tool for leveraging large-scale unlabeled data, addressing data scarcity and improving model robustness [18] [64] [65].

5. Predictive Accuracy and Clinical Applicability

Current Predictive Performance

- **RNNs:** Achieve high accuracy (up to 98.8%) in disease prediction tasks, outperforming traditional models [23] [66] [67].
- **Hybrid Models:** CNN-RNN and Bayesian neural networks provide robust predictions and uncertainty quantification [4] [68].
- **Traditional ML:** Logistic Regression and SVM remain competitive, especially in interpretability and clinical settings [69] [70].

Synthesis: Neural network models demonstrate high predictive accuracy, but clinical applicability depends on interpretability, external validation, and integration into workflows [23] [66] [67].

Clinical Translation: Barriers and Opportunities

- **Barriers:** Lack of external validation, interpretability, data heterogeneity, and integration challenges limit clinical adoption [71] [72] [73].
- **Opportunities:** Robust external validation, standardized guidelines, and interpretable systems are needed for clinical readiness [71] [72].

Synthesis: Overcoming barriers to clinical translation requires rigorous validation, transparent models, and seamless integration with clinical systems [71] [72] [73].

Uncertainty Quantification and Decision Support

- **Bayesian Neural Networks:** Provide confidence estimates and risk stratification, supporting clinical decision-making [13] [14] [74].
- **Ensemble Methods:** Enhance reliability and robustness in predictions [75] [76].

Synthesis: Uncertainty quantification is essential for clinical trust, enabling risk-based decision support and early identification of emerging variants [13] [14] [74].

Interpretability Techniques for Clinical Trust

- **SHAP, LIME, Grad-CAM:** Improve transparency and feature attribution, fostering clinical trust [10] [77] [78].
- **Rule-Based Frameworks:** Extract symbolic rules for decision support [11].
- **Human-Centered Design:** Engages clinicians in model development and explanation [79].

Synthesis: Interpretability techniques are critical for clinical adoption, aligning model explanations with medical expertise and supporting actionable insights [10] [77] [78].

6. Best Practices, Guidelines, and Future Directions

Best Practices for Model Development and Validation

- **Data Preprocessing:** Standardize and denoise genomic data for improved model accuracy [80] [81] [82].
- **Feature Selection:** Use methods like Chi-Square and Mutual Information to enhance performance and reduce complexity [80].
- **Interpretability:** Integrate explainable AI techniques for transparent predictions [83].
- **Handling Data Imbalance:** Employ SMOTE and advanced sampling methods [84].

Synthesis: Adhering to best practices in data handling, feature selection, and interpretability is essential for robust and clinically relevant models [80] [81] [82].

Standardized Guidelines and Regulatory Considerations

- **Explainable AI:** Mechanistic insights and transparency are increasingly emphasized [85].
- **Regulatory Needs:** Standardized protocols for model construction, validation, and interpretation are lacking [53] [85] [86].
- **Ethical Compliance:** Responsible data usage and privacy-preserving methods are critical [53].

Synthesis: The field requires universally accepted guidelines and regulatory frameworks to ensure safe, effective, and equitable clinical translation [53] [85] [86].

Emerging Trends and Research Gaps

- **Multimodal Integration:** Combining genomic, clinical, and imaging data for comprehensive disease prediction [29] [87].
- **Explainable AI:** Advancing interpretability for actionable clinical insights [88].
- **Federated Learning:** Privacy-preserving collaborative training on sensitive data [19].
- **Self-Supervised Learning:** Leveraging unlabeled data for improved accuracy and generalization [18].

Research Gaps: Standardized guidelines, external validation, interpretability, and integration of multi-omics and clinical data remain underdeveloped [29] [87] [88].

7. Conclusion

Summary and Outlook

Neural network models for DNA sequence-based disease prediction are advancing rapidly, with hybrid architectures, transformer-based genome language models, and NAS-optimized frameworks setting new standards in predictive accuracy and interpretability. The integration of multi-modal data, robust uncertainty quantification, and explainable AI techniques is essential for clinical translation. However, challenges in data quality, external validation, regulatory compliance, and interpretability persist. Addressing these gaps through standardized guidelines, modular pipelines, and clinician engagement will pave the way for actionable, trustworthy, and personalized healthcare solutions [17] [23] [89].

Creative Insight: Developing a modular pipeline that predicts disease risk from DNA sequences and provides interpretable reports highlighting influential genetic markers, coupled with user-friendly visualization tools and uncertainty quantification, can make predictions actionable for healthcare professionals.

Further Reading:

- Reviews on k-mer vs. transformer-based embeddings [1]
- Hybrid GNN-GAN models [5]
- Self-supervised learning in genomics [18]
- Multi-omics integration surveys [6] [21]

End of Report

References

1. GSRNet, an adversarial training-based deep framework with multi-scale CNN and BiGRU for predicting genomic signals and regions Zhu, G., Fan, Y., Li, F., (...), Zhou, F. Expert Systems with Applications, 2023 https://www.scopus.com/pages/publications/85160207822?origin=scopusAI

2. Gene-LLMs: a comprehensive survey of transformer-based genomic language models for regulatory and clinical genomics Balakrishnan, P., Anny Leema, A., Dhivya Shree, V., (...), Mohan Babu, A. Frontiers in Genetics, 2025 https://www.scopus.com/pages/publications/105019960352?origin=scopusAI

3. A 16nm Fully Integrated SoC for Hardware-Aware Neural Architecture Search Lin, Y.-C., Huang, M.-S., Wang, J.-B., (...), Yang, C.-H. European Solid-State Circuits Conference, 2025 https://www.scopus.com/pages/publications/105024546249?origin=scopusAI

4. Investigation of Disease Prediction Based on DNA Using Deep Model Techniques Puvaneswari, S., Indirani, G., Revathy, G. AI Techniques for Association Rule Mining in Medical Data: Trends and Practical Applications, 2026 https://www.scopus.com/pages/publications/105040015096?origin=scopusAI

5. Applications of generative adversarial networks in neuroimaging and clinical neuroscience Wang, R., Bashyam, V., Yang, Z., (...), Davatzikos, C. NeuroImage, 2023 https://www.scopus.com/pages/publications/85147205607?origin=scopusAI

6. Multi-Modal Disease Prediction With Hierarchical Self-Supervised Learning Qu, Z., Chen, T., Zhou, X., (...), Cui, L. IEEE Journal of Biomedical and Health Informatics, 2025 https://www.scopus.com/pages/publications/105003080946?origin=scopusAI

7. Multimodal graph neural networks for integrating heterogeneous medical data in clinical decision making Reddy, K.A.N., Lakshmi, L., Prasad, G.S.C., (...), Tilaye, G.F. Discover Computing, 2026 https://www.scopus.com/pages/publications/105048136190?origin=scopusAI

8. Hybrid Parallel Fuzzy CNN Paradigm: Unmasking Intricacies for Accurate Brain MRI Insights Iqbal, S., Qureshi, A.N., Aurangzeb, K., (...), Khan, F. IEEE Transactions on Fuzzy Systems, 2024 https://www.scopus.com/pages/publications/85187380151?origin=scopusAI

9. Deep Learning and Explainable AI: New Pathways to Genetic Insights Wang, C., Zuo, C., Su, Z., (...), Zhang, Z. Frontiers in Artificial Intelligence and Applications, 2025 https://www.scopus.com/pages/publications/105024408610?origin=scopusAI

10. A Comparative Analysis of Explainable AI Techniques for Enhanced Model Interpretability Swathi, Y., Challa, M. Proceedings - 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN 2023, 2023 https://www.scopus.com/pages/publications/85175240267?origin=scopusAI

11. Systematic Literature Review in Rule-based Interpretable Deep Neural Networks Chandrinos, N., Chatzistamatis, S., Tsekouras, G.E. 2026 6th International Conference on Innovative Research in Applied Science, Engineering and Technology, IRASET 2026, 2026 https://www.scopus.com/pages/publications/105042529885?origin=scopusAI

12. Interpreting Deep Convolutional Neural Networks for Medical Diagnosis Using Perceptual Encoding and Retrieval Kasture, S., Maurya, S., Verma, R., (...), Shukla, A. 2024 15th International Conference on Computing Communication and Networking Technologies, ICCCNT 2024, 2024 https://www.scopus.com/pages/publications/85211120218?origin=scopusAI

13. Bayesian networks for predicting clinical outcomes in COVID-19 patients: A retrospective study in a resource-limited setting Filamant, T.C., Raherinirina, A.F., Totohasina, A. PLOS ONE, 2026 https://www.scopus.com/pages/publications/105032671310?origin=scopusAI

14. Uncertainty quantification using Bayesian neural networks in classification: Application to biomedical image segmentation Kwon, Y., Won, J.-H., Kim, B.J., Paik, M.C. Computational Statistics and Data Analysis, 2020 https://www.scopus.com/pages/publications/85069976043?origin=scopusAI

15. A Simple Approach to Improve Single-Model Deep Uncertainty via Distance-Awareness Liu, J.Z., Padhy, S., Ren, J., (...), Lakshminarayanan, B. Journal of Machine Learning Research, 2023 https://www.scopus.com/pages/publications/85213836448?origin=scopusAI

16. Exploring the performance of genomic prediction models for soybean yield using different validation approaches Đorđević, V., Ćeran, M., Miladinović, J., (...), Marinković, J. Molecular Breeding, 2019 https://www.scopus.com/pages/publications/85065741333?origin=scopusAI

17. Artificial Intelligence Approaches in Genomic Disease Prediction Fakir, W., Fakir, Y. Communications in Computer and Information Science, 2026 https://www.scopus.com/pages/publications/105030246881?origin=scopusAI

18. A self-supervised deep learning method for data-efficient training in genomics Gündüz, H.A., Binder, M., To, X.-Y., (...), Rezaei, M. Communications Biology, 2023 https://www.scopus.com/pages/publications/85170632134?origin=scopusAI

19. MQFL-FHE: Multimodal Quantum Federated Learning Framework with Fully Homomorphic Encryption Dutta, S., Innan, N., Yahia, S.B., (...), Bernal Neira, D.E. Proceedings of the International Joint Conference on Neural Networks, 2025 https://www.scopus.com/pages/publications/105029249854?origin=scopusAI

20. Hy-GeneNet: A Hybrid Ensemble of N-Gram and Attention-Based LSTM for Reliable and Interpretable DNA Gene Sequence Prediction Alam, T., Rahman, S., Al Jubair, A., Hossen, M.J. 2025 28th International Conference on Computer and Information Technology, ICCIT 2025, 2025 https://www.scopus.com/pages/publications/105041660866?origin=scopusAI

21. SFGAE: A self-feature-based graph auto encoder model for miRNA-disease associations prediction Ma, M., Na, S., Zhang, X., (...), Xu, J. Briefings in Bioinformatics, 2022 https://www.scopus.com/pages/publications/85138492241?origin=scopusAI

22. Neural Network Analysis Joshi, A., Sasumana, J., Ray, N.M., Kaushik, V. Advances in Bioinformatics, 2021 https://www.scopus.com/pages/publications/85145348534?origin=scopusAI

23. Towards a robust out-of-the-box neural network model for genomic data Zhang, Z., Cheng, S., Solis-Lemus, C. BMC Bioinformatics, 2022 https://www.scopus.com/pages/publications/85127903199?origin=scopusAI

24. An Efficient Deep Convolutional Neural Networks Model for Genomic Sequence Classification Pimpalkar, A., Gandhewar, N., Shelke, N., (...), Chhabria, S. Genomics at the Nexus of AI, Computer Vision, and Machine

Learning, 2024 https://www.scopus.com/pages/publications/85209838696?origin=scopusAI

25. Deep Pulse: Exploring Bi-LSTM, GNN and GAN Models for Heart Disease Prediction Sunmesh, S., Mohammed, N., Srivastava, H., (...), Jasmine Selvakumari Jeya, I. Proceedings - 2025 7th International Conference on Artificial Intelligence and Speech Technology, AIST 2025, 2025 https://www.scopus.com/pages/publications/105037191981?origin=scopusAI

26. Comprehensive evaluation of deep learning architectures for prediction of DNA/RNA sequence binding specificities Trabelsi, A., Chaabane, M., Ben-Hur, A. Bioinformatics, 2019 https://www.scopus.com/pages/publications/85068915583?origin=scopusAI

27. ADVANCED DISEASE DETECTION USING HYBRID CNN WITH LSTM AND GRU MODELS: A DEEP LEARNING APPROACH Jadhav, N., Makandar, A. Journal of Theoretical and Applied Information Technology, 2025 https://www.scopus.com/pages/publications/85216317201?origin=scopusAI

28. Classification of Electrocardiogram of Congenital Heart Disease Patients by Neural Network Algorithms Yuan, Y., Zhang, Y., Wang, J., Fang, P. Scientific Programming, 2021 https://www.scopus.com/pages/publications/85115016634?origin=scopusAI

29. Multimodal foundation transformer models for multiscale genomics Khan, S.A., Martínez-de-Morentin, X., Alsabbagh, A.R., (...), Tegner, J. Nature Methods, 2026 https://www.scopus.com/pages/publications/105023554218?origin=scopusAI

30. Protein Function Prediction with Pretrained Transformers: Performance, Pitfalls, and Practical Guidance Roy, K.R. Eurobiotech Journal, 2026 https://www.scopus.com/pages/publications/105038421967?origin=scopusAI

31. Template-based graph registration network for boosting the diagnosis of brain connectivity disorders Gürler, Z., Gharsallaoui, M.A., Rekik, I. Computerized Medical Imaging and Graphics, 2023 https://www.scopus.com/pages/publications/85143542458?origin=scopusAI

32. Graph Neural Networks for Medical Imaging Analysis and Biological Data: Integrating Topology, Geometry, Radiomics, and Generative AI Singh, Y., Himeur, Y., Farrelly, C.M., (...), Hathaway, Q.A. Bioengineering, 2026 https://www.scopus.com/pages/publications/105042845369?origin=scopusAI

33. BANANAS: Bayesian Optimization with Neural Architectures for Neural Architecture Search White, C., Neiswanger, W., Savani, Y. 35th AAAI Conference on Artificial Intelligence, AAAI 2021, 2021 https://www.scopus.com/pages/publications/85130067302?origin=scopusAI

34. Training-free neural architecture search: A review Wu, M.-T., Tsai, C.-W. ICT Express, 2024 https://www.scopus.com/pages/publications/85177220724?origin=scopusAI

35. Semi-supervised accuracy predictor-based multi-objective neural architecture search Xiao, S., Zhao, B., Liu, D. Neurocomputing, 2024 https://www.scopus.com/pages/publications/85202914815?origin=scopusAI

36. Enhancing multi-objective evolutionary neural architecture search with training-free Pareto local search Phan, Q.M., Luong, N.H. Applied Intelligence, 2023 https://www.scopus.com/pages/publications/85136968090?origin=scopusAI

37. A novel explainable deep-learning approach for network analysis of epistatic interactions Mastropietro, A., Markopoulos, G., Evangelou, E., Anagnostopoulos, A. NAR Genomics and Bioinformatics, 2026 https://www.scopus.com/pages/publications/105028917062?origin=scopusAI

38. Interpreting High Order Epistasis Using Sparse Transformers Graca, M., Marques, D., Santander-Jimenez, S., (...), Ilic, A. Proceedings - 2023 IEEE/ACM International Conference on Connected Health: Applications, Systems and Engineering Technologies, CHASE 2023, 2023 https://www.scopus.com/pages/publications/85167409256?origin=scopusAI

39. GPBSO: Gene Pool-Based Brain Storm Optimization for SNP Epistasis Detection Sun, L., Xin, Y., Qu, S., (...), Jiang, L. Genes, 2025 https://www.scopus.com/pages/publications/105017413280?origin=scopusAI

40. Epi-SSA: A novel epistasis detection method based on a multi-objective sparrow search algorithm Sun, L., Bian, J., Xin, Y., (...), Zheng, L. PLoS ONE, 2024 https://www.scopus.com/pages/publications/85207361237?origin=scopusAI

41. A Review of Ant Colony Optimization Based Methods for Detecting Epistatic Interactions Shang, J., Wang, X., Wu, X., (...), Zhang, H. IEEE Access, 2019 https://www.scopus.com/pages/publications/85061743425?origin=scopusAI

42. EpiRL: A Reinforcement Learning Agent to Facilitate Epistasis Detection Huang, K., Nogueira, R. Studies in Computational Intelligence, 2020 https://www.scopus.com/pages/publications/85070536254?origin=scopusAI

43. Enhancing disease prediction on imbalanced metagenomic dataset by cost-sensitive Nguyen, H.T., Tran, T.B., Bui, Q.M., (...), Tran, N.C. International Journal of Advanced Computer Science and Applications, 2020 https://www.scopus.com/pages/publications/85088955895?origin=scopusAI

44. A novel multi-modal dual pathway network with hierarchical channel-spatial attention and adaptive feature fusion for viral genomic variant classification Fadia, L., Shah, V., Hassanzadeh, M., (...), Ahmadi, M. Network Modeling Analysis in Health Informatics and Bioinformatics, 2025 https://www.scopus.com/pages/publications/105012300761?origin=scopusAI

45. Deep gate information bottleneck-based prediction model for complex disease-related micro-ribonucleic acids via heterogeneous biological networks Guo, Y., Xin, Y., Cao, J., (...), Zhou, D. Engineering Applications of Artificial Intelligence, 2025 https://www.scopus.com/pages/publications/105005518473?origin=scopusAI

46. Graph Learning Approaches for Graph with Noise: Application to Disease Prediction in Population Graph Chen, L., Huang, Y., Liao, B., (...), Hu, J. Proceedings - 2020 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2020, 2020 https://www.scopus.com/pages/publications/85100354716?origin=scopusAI

47. Chex-Mix: Combining Homomorphic Encryption with Trusted Execution Environments for Oblivious Inference in the Cloud Natarajan, D., Loveless, A., Dai, W., Dreslinski, R. Proceedings - 8th IEEE European Symposium on Security and Privacy, Euro S and P 2023, 2023 https://www.scopus.com/pages/publications/85168095396?origin=scopusAI

48. Interpretation of deep learning in genomics and epigenomics Talukder, A., Barham, C., Li, X., Hu, H. Briefings in Bioinformatics, 2021 https://www.scopus.com/pages/publications/85106715368?origin=scopusAI

49. Interpretable Multi-Task Learning Framework for DNA Intelligence Bhumika, C., Nayak, P., Janapati, R., (...), Trivedi, V. OCIT 2025 - Proceedings 23rd OITS International Conference on Information Technology, 2025 https://www.scopus.com/pages/publications/105036726927?origin=scopusAI

50. Deep GONet: self-explainable deep neural network based on Gene Ontology for phenotype prediction from gene expression data Bourgeais, V., Zehraoui, F., Ben Hamdoune, M., Hanczar, B. BMC Bioinformatics, 2021 https://www.scopus.com/pages/publications/85115254163?origin=scopusAI

51. A systematic review of biologically-informed deep learning models for cancer: fundamental trends for encoding and interpreting oncology data Wysocka, M., Wysocki, O., Zufferey, M., (...), Freitas, A. BMC Bioinformatics, 2023 https://www.scopus.com/pages/publications/85159422811?origin=scopusAI

52. ExplaiNN: interpretable and transparent neural networks for genomics Novakovsky, G., Fornes, O., Saraswat, M., (...), Wasserman, W.W. Genome Biology, 2023 https://www.scopus.com/pages/publications/85163783542?origin=scopusAI

53. Artificial intelligence and its application in clinical microbiology Mairi, A., Hamza, L., Touati, A. Expert Review of Anti-Infective Therapy, 2025 https://www.scopus.com/pages/publications/105001121385?origin=scopusAI

54. Using core genome and machine learning for serovar prediction in Salmonella enterica subspecies I strains Li, X., Oladeinde, A., Rothrock, M., (...), Ghazi Al Hakeem, W. FEMS Microbiology Letters, 2025 https://www.scopus.com/pages/publications/105003284079?origin=scopusAI

55. Human miRNA–disease Association Prediction Via Residual GraphSAGE With Nonlinear Adaptive Feature Fusion and Triplet Contrastive Learning Sui, J., Cui, W., Zhang, X., (...), Guo, J. Journal of Molecular Biology, 2025 https://www.scopus.com/pages/publications/105012156162?origin=scopusAI

56. Deep learning on graphs for multi-omics classification of COPD Zhuang, Y., Xing, F., Ghosh, D., (...), Kechris, K. PLoS ONE, 2023 https://www.scopus.com/pages/publications/85153551769?origin=scopusAI

57. GPLMD: A Multi-Task Graph Learning Framework for inferring the relationships among lncRNAs, miRNAs and diseases Sun, R., Chen, X., Zhao, D., (...), Cai, S. Computers in Biology and Medicine, 2025 https://www.scopus.com/pages/publications/105012629510?origin=scopusAI

58. An Empirical Analysis of Transformer-Based and Convolutional Neural Network Approaches for Early Detection and Diagnosis of Cancer Using Multimodal Imaging and Genomic Data Sangeetha, S.K.B., Kumar Mathivanan, S., Muthukumaran, V., (...), Easwaramoorthy, S.V. IEEE Access, 2025 https://www.scopus.com/pages/publications/85214132333?origin=scopusAI

59. Advanced Heart Disease Prediction Using Fuzzy-Rough Sets and Enhanced Missing Data Imputation Techniques Cenitta, D., Shravya, A.R., Madli, R., (...), Vijaya Arjunan, R. Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications, 2025 https://www.scopus.com/pages/publications/105003063063?origin=scopusAI

60. Benchmarking interpretability of deep learning for predictive genomics: Recall, precision, and variability of feature attribution Reynolds, J., Pan, C. PLOS Computational Biology, 2025 https://www.scopus.com/pages/publications/105024040741?origin=scopusAI

61. Integrating Convolutional Neural Networks for Genomic Sequence Analysis: Deep Learning Applications in Genomics and Bioinformatics Pawar, S., Kanika, Srividhya, S., (...), Mohana Priya, T. 3rd International Conference on Advances in Computing, Communication and Materials, ICACCM 2024, 2024 https://www.scopus.com/pages/publications/105011268677?origin=scopusAI

62. Combining Neuroimaging and Omics Datasets for Disease Classification Using Graph Neural Networks Chan, Y.H., Wang, C., Soh, W.K., Rajapakse, J.C. Frontiers in Neuroscience, 2022 https://www.scopus.com/pages/publications/85131787985?origin=scopusAI

63. Multi-Modal Deep Learning Approach for Merging Genomic and Clinical Data in Cancer Prognosis Shah, D.A., Shaikh, U.H., Jha, S., (...), Verma, S. 2025 3rd DMIHER International Conference on Artificial Intelligence in Healthcare, Education and Industry, IDICAIHEI 2025, 2025 https://www.scopus.com/pages/publications/105034715373?origin=scopusAI

64. Systematic comparison of semi-supervised and self-supervised learning for medical image classification Huang, Z., Jiang, R., Aeron, S., Hughes, M.C. Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2024 https://www.scopus.com/pages/publications/85204111854?origin=scopusAI

65. Semi-supervised learning improves gene expression-based prediction of cancer recurrence Shi, M., Zhang, B. Bioinformatics, 2011 https://www.scopus.com/pages/publications/80054895643?origin=scopusAI

66. A Comparitive Study of RNN and Restricted BOLTZMANN MACHINES(RBM) in Predicting HIV Test Results Analyzing Using The EHDS Dataset Bash, S.M.D.H., Puviarasi, R. 2025 10th International Conference on Applying New Technology in Green Buildings, ATiGB 2025, 2025 https://www.scopus.com/pages/publications/105017127960?origin=scopusAI

67. The Role of Mathematics in Machine Learning for Disease Prediction: An In-Depth Review in the Healthcare Domain Abdillah, Syaharuddin, Mandailina, V., Mehmood, S. Advance Sustainable Science, Engineering and Technology, 2024 https://www.scopus.com/pages/publications/85209177979?origin=scopusAI

68. Application of Bayesian Neural Networks in Healthcare: Three Case Studies Ngartera, L., Issaka, M.A., Nadarajah, S. Machine Learning and Knowledge Extraction, 2024 https://www.scopus.com/pages/publications/85213474006?origin=scopusAI

69. Machine Learning Algorithms for Disease Prediction: A Comparative Analysis of Conventional and Advanced Approaches Balachander, M., Jubilson, E.A., Natarajan, K. Information Systems Engineering and Management, 2026 https://www.scopus.com/pages/publications/105048640312?origin=scopusAI

70. Predicting Heart Disease with Machine Learning: A Comparative Analysis of Multiple Models Afroz, S., Islam, M.M., Sharma, N., (...), Ali, M.A. 2025 5th International Conference on Emerging Research in Electronics, Computer Science and Technology, ICERECT 2025, 2025 https://www.scopus.com/pages/publications/105034865498?origin=scopusAI

71. A systematic review of machine learning in heart disease prediction Banerjee, T., Paçal, İ. Turkish Journal of Biology, 2025 https://www.scopus.com/pages/publications/105022239026?origin=scopusAI

72. External validation of a knowledge-based clinical toxicology consultant for diagnosing single exposures Lerogeron, H., Schipper, J., Schauben, J., (...), Nguyen, K.A. Clinical Toxicology, 2026 https://www.scopus.com/pages/publications/105019659634?origin=scopusAI

73. Toward Implementation of Artificial Neural Networks That "Really Work" Leon, M.A., Keller, J. Journal of the American Medical Informatics Association, 1997 https://www.scopus.com/pages/publications/0347840400?origin=scopusAI

74. Uncertainty-weighted learning to reduce diagnostic uncertainty, evaluate AI models, and augment precise clinical decisions Li, J., Arefan, D., Wu, S. Progress in Biomedical Optics and Imaging - Proceedings of SPIE, 2026 https://www.scopus.com/pages/publications/105041070426?origin=scopusAI

75. A general and efficient approach for uncertainty quantification in neural networks: Identifying risky decisions in AI systems Zhang, Z., Luo, S., Wu, X., (...), Pan, L. Advanced Engineering Informatics, 2026 https://www.scopus.com/pages/publications/105028089204?origin=scopusAI

76. Single-model uncertainty quantification in neural network potentials does not consistently outperform model ensembles Tan, A.R., Urata, S., Goldman, S., (...), Gómez-Bombarelli, R. npj Computational Materials, 2023 https://www.scopus.com/pages/publications/85179847141?origin=scopusAI

77. Explainability techniques for Artificial Intelligence models in medical diagnostic Falvo, F.R., Cannataro, M. Proceedings - 2024 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2024, 2024 https://www.scopus.com/pages/publications/85217275594?origin=scopusAI

78. ARTIFICIAL INTELLIGENCE AND NEURAL NETWORK INTERPRETATION TECHNIQUES FOR IMAGE VISUALIZATION AND RECOGNITION Zhang, R., Wang, B., Liu, J., Liu, Y. Journal of Mechanics in Medicine and Biology, 2026 https://www.scopus.com/pages/publications/105034580324?origin=scopusAI

79. A Human-Centered Approach to Interpretable Machine Learning in Clinical Decision Support Systems Nimbalkar, V., Chawla, L., Adnan, M.M., (...), Kalra, R. 2nd International Conference on Artificial Intelligence for

Innovations in Healthcare Industries, ICAIIHI 2025, 2025
https://www.scopus.com/pages/publications/105041770175?origin=scopusAI

80. An Efficient Genetic Disorder Classification Framework Using Feature Selection, Machine Learning and Deep Learning Techniques Sony, R., Manike, C. Proceedings of 7th International Conference on Smart Systems and Inventive Technology, ICSSIT 2026, 2026 https://www.scopus.com/pages/publications/105049043279?origin=scopusAI

81. Integrating Multi-omics and Clinical Narratives for Predictive Modeling: Genomics, Transcriptomics, Proteomics, and Medical Texts in Disease Analysis Irfan, S., Ali, R.N., Ali, R.F., Hassan, B. Smart Innovation, Systems and Technologies, 2025 https://www.scopus.com/pages/publications/105000812669?origin=scopusAI

82. Tabular deep learning: a comparative study applied to multi-task genome-wide prediction Fan, Y., Waldmann, P. BMC Bioinformatics, 2024 https://www.scopus.com/pages/publications/85205826603?origin=scopusAI

83. A Deep Learning and Explainable AI-Based Approach for the Classification of Discomycetes Species Korkmaz, A.F., Ekinci, F., Altaş, Ş., (...), Akata, I. Biology, 2025
https://www.scopus.com/pages/publications/105011749997?origin=scopusAI

84. Advanced Feature Engineering and Sampling Technique for Effective Heart Disease Prediction System using Machine Learning Technique Awoyemi, J.O., Awoyemi, B.O., Ugwu, T.A., (...), Ugwu, C.C. IEEE International Conference on Emerging and Sustainable Technologies for Power and ICT in a Developing Society, NIGERCON, 2024 https://www.scopus.com/pages/publications/105005937549?origin=scopusAI

85. Obtaining genetics insights from deep learning via explainable artificial intelligence Novakovsky, G., Dexter, N., Libbrecht, M.W., (...), Mostafavi, S. Nature Reviews Genetics, 2023
https://www.scopus.com/pages/publications/85139658123?origin=scopusAI

86. Focusing on AI hematology prediction model and paying close attention to process quality and efficiency Qu, L., Xu, W. Chinese Journal of Laboratory Medicine, 2025
https://www.scopus.com/pages/publications/105005280366?origin=scopusAI

87. Designing Advanced Algorithmic Frameworks and Deep Architectures for Self-Supervised Representation Learning in Medical Imaging Asaad, R.R., Almufti, S.M., Ali, R.I., (...), Nasir, B.O. GSEACT 2026 - 2026 IEEE Global Symposium on Emerging and Communication Technologies, 2026
https://www.scopus.com/pages/publications/105047876952?origin=scopusAI

88. Deep Learning and Neural Networks: Unlocking Complex Patterns in Genomic Medicine Sudharsan, M., Maithili, K., Ravi, T., (...), Eswaran, P. Pharmacogenomics Using Artificial Intelligence: Optimizing Drug Response through Personalized Genomic Analysis, 2026
https://www.scopus.com/pages/publications/105043774070?origin=scopusAI

89. Transformers and genome language models Consens, M.E., Dufault, C., Wainberg, M., (...), Wang, B. Nature Machine Intelligence, 2025 https://www.scopus.com/pages/publications/105000075914?origin=scopusAI